

RetentionIQ: Telecom Churn Intelligence

Project Report

Total Customers	Churn Rate	Churned	New Joiners	Predicted Churners
6,418	27.0%	1,732	411	380

Project purpose

Build a end-to-end churn analytics solution that starts with SQL Server preprocessing, continues into Power BI storytelling, and adds a Random Forest model to predict future churn risk.

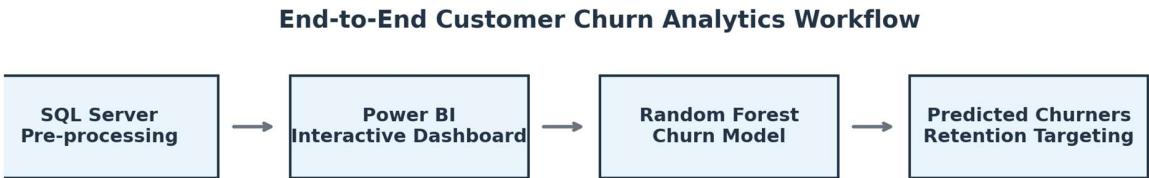


Figure 1. End-to-end workflow from SQL Server to Power BI and ML predictions.

Recommended repository name	retentioniq-telecom-churn-intelligence
Primary tools	SQL Server, Power BI, Python, pandas, scikit-learn
Dataset snapshot	Customer records, service subscriptions, charges, churn status, churn reasons, and predicted churners
Business outcome	Identify churn drivers and prioritize retention outreach

Contents

1. Executive Summary
2. Business Problem and Objectives
3. Data Overview
4. SQL Server Preparation Layer
5. Power BI Dashboard Design
6. Exploratory Insights
7. Random Forest Churn Prediction Model
8. Predicted Churner Prioritization
9. Business Recommendations
10. GitHub Presentation and Resume Positioning
11. Limitations and Next Steps
12. Appendix: SQL Business Questions

1. Executive Summary

This project analyzes customer churn for a telecom-style subscription business. The solution combines SQL Server data preparation, Power BI dashboards, and a Random Forest model to move from descriptive churn reporting to proactive churn prevention. The portfolio contains 6,418 customers, including 1,732 churned customers and 411 new joiners. The overall churn rate is 27.0% based on the provided project snapshot.

- Churn is concentrated among month-to-month contracts, customers over 50, selected high-risk states, and non-card payment methods.
- Competitor-related churn is the largest churn category, indicating that pricing, devices, speed, and data plan competitiveness require attention.
- The Random Forest model achieved 84% holdout accuracy and identified 380 predicted churners for retention targeting.
- The SQL layer includes reusable feature engineering, data quality checks, advanced segmentation, cohorting, and model-output prioritization queries.

Business value

The project gives stakeholders an actionable retention workflow: diagnose churn drivers, monitor them in Power BI, and prioritize predicted churners for intervention.

2. Business Problem and Objectives

Recurring-revenue businesses need to identify customers at risk before they leave. The goal of this project is to provide a single analytics workflow that answers what is happening, why it is happening, and which customers should be contacted first.

Initial business questions

1. What is the overall churn rate and how many customers are stayed, churned, or newly joined?
2. Which customer segments have the highest churn rate by contract, payment method, tenure, age group, state, and internet type?
3. Which churn reasons and churn categories contribute the highest churn volume?
4. Which services are most common among churned customers?
5. Which new joiners are predicted to churn and should be prioritized for retention campaigns?
6. What revenue exposure is associated with churned and predicted churn customers?

Success criteria

- Build a clean, explainable SQL preparation layer.

- Create Power BI dashboards that communicate executive KPIs and operational retention targets.
- Train and evaluate a machine learning model that predicts churn for future customers.
- Package the work so recruiters can quickly understand the business impact and technical depth.

3. Data Overview

Metric	Value
Total customer rows	6,418
Stayed customers	4,275
Churned customers	1,732
New joiners	411
Overall churn rate	27.0%
Historical churn rate excluding new joiners	28.8%
Total revenue from churned customers	3,411,960.58
Predicted churners in model output	380

Important field groups

Field group	Example columns	Analytical use
Customer profile	Customer_ID, Gender, Age, Married, State	Segment churn risk and build retention personas
Relationship history	Tenure_in_Months, Number_of_Referrals, Contract	Understand loyalty and commitment patterns
Services	Internet_Service, Online_Security, Premium_Support, Streaming_TV	Compare service adoption with churn behavior
Billing and value	Monthly_Charge, Total_Charges, Refunds, Total_Revenue	Estimate revenue exposure and value-based prioritization
Churn labels	Customer_Status, Churn_Category, Churn_Reason	Train ML model and explain churn reasons

4. SQL Server Preparation Layer

The SQL layer is designed as a reusable analytics foundation. It standardizes blank values, creates churn flags, groups age and tenure into business-friendly bands, builds monthly charge bands, and adds data quality flags. This makes Power BI visuals and ML feature preparation more consistent.

SQL techniques demonstrated

- Common Table Expressions for clean analytical steps.
- Window functions such as ROW_NUMBER, RANK, DENSE_RANK, NTILE, and cumulative SUM OVER.
- Segment-level churn lift calculations against portfolio churn rate.
- CROSS APPLY service unpivoting to analyze multiple service columns in one query.
- Pareto analysis for churn reasons and revenue exposure.
- Data quality checks for negative charges, missing keys, and invalid statuses.

Deliverable

The SQL file is provided as sql/customer_churn_business_questions.sql and can be executed in SQL Server after importing the CSV into dbo.Customer_Data.

5. Power BI Dashboard Design

The Power BI report contains two pages. The first page explains current churn performance and churn drivers. The second page focuses on the prediction output and lists customers predicted to churn.

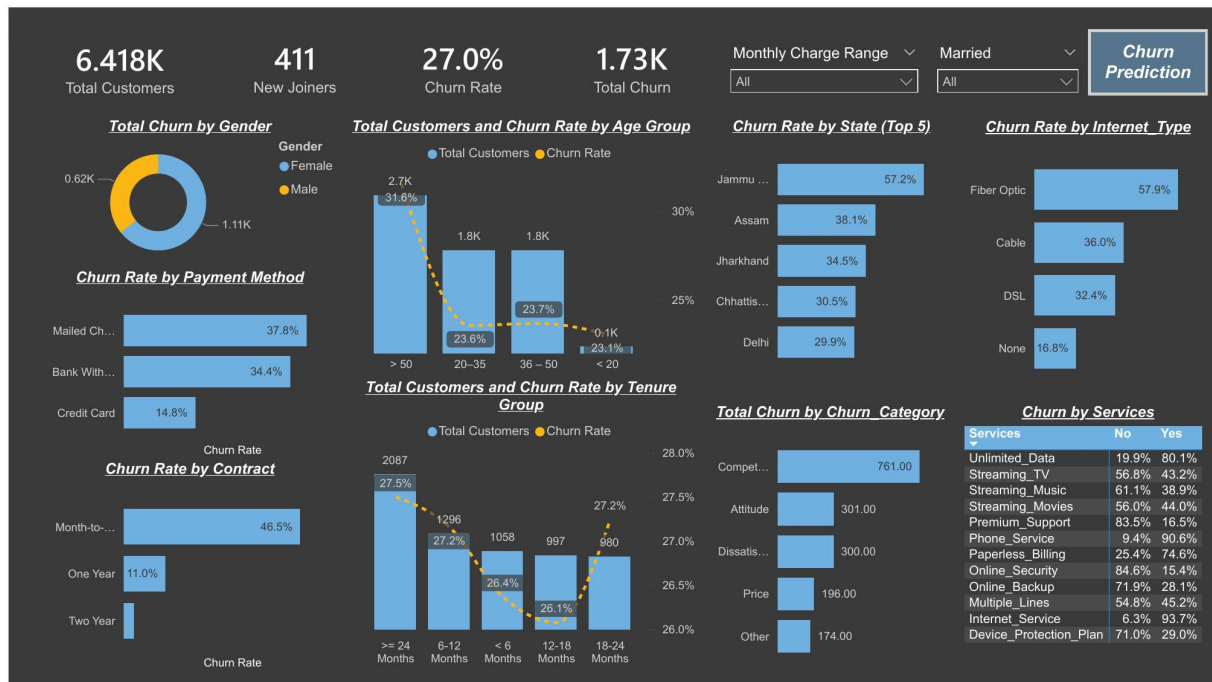


Figure 2. Power BI churn analysis dashboard.

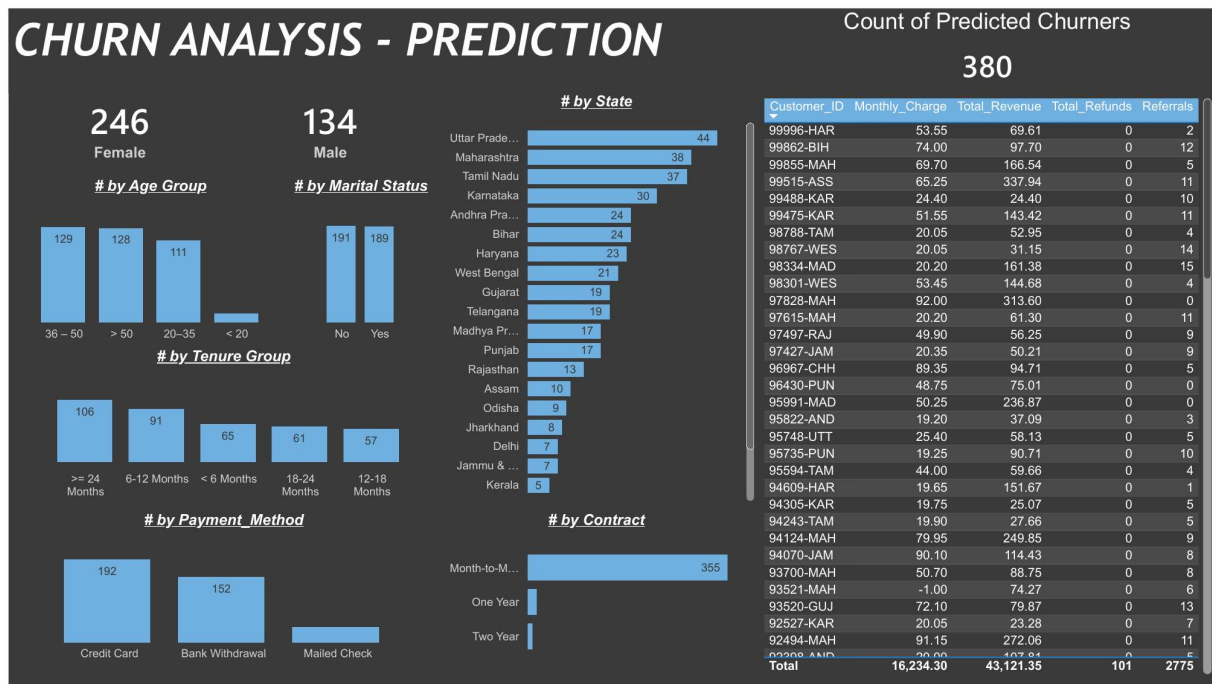


Figure 3. Power BI churn prediction dashboard.

Dashboard design strengths

- Executive KPI cards show total customers, churn rate, churned customers, and new joiners.
- Segment charts explain churn by gender, age group, state, contract, payment method, internet type, services, and tenure.
- Prediction page converts model output into a retention call list with customer-level monthly charge, revenue, refunds, and referrals.
- Slicers allow stakeholders to filter by monthly charge range and marital status.

6. Exploratory Insights

Contract churn is the strongest visible lever

Month-to-month customers churn at 46.5%, far above one-year and two-year contract groups. This makes contract conversion one of the most actionable retention strategies.

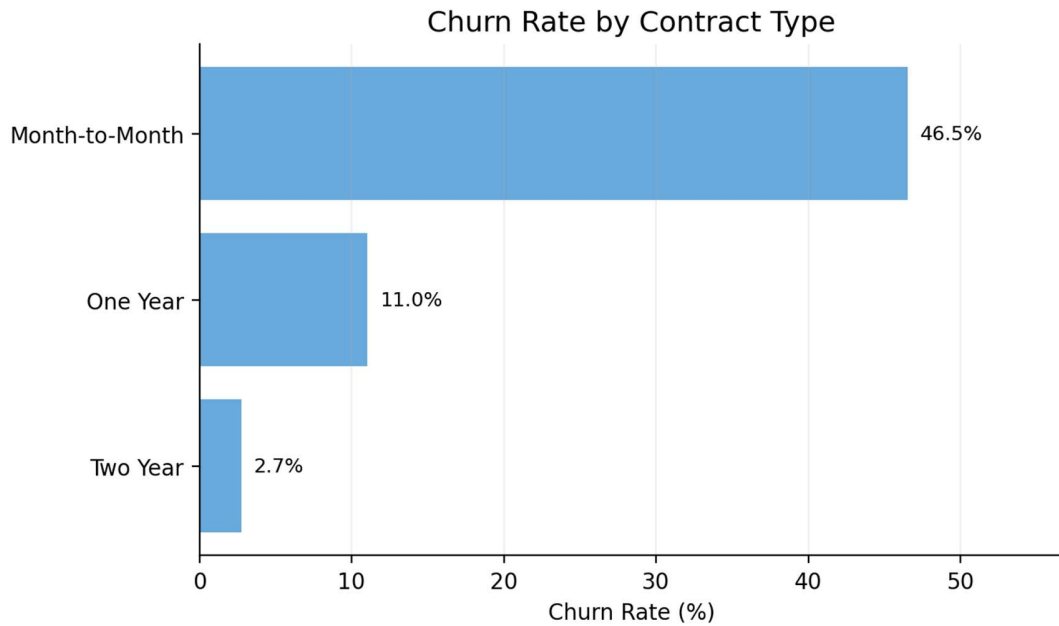


Figure 4. Churn rate by contract type.

Geographic risk is concentrated

Churn rate varies materially by state. Jammu & Kashmir, Assam, Jharkhand, Chhattisgarh, and Delhi appear as priority states in the dashboard and supporting SQL analysis.

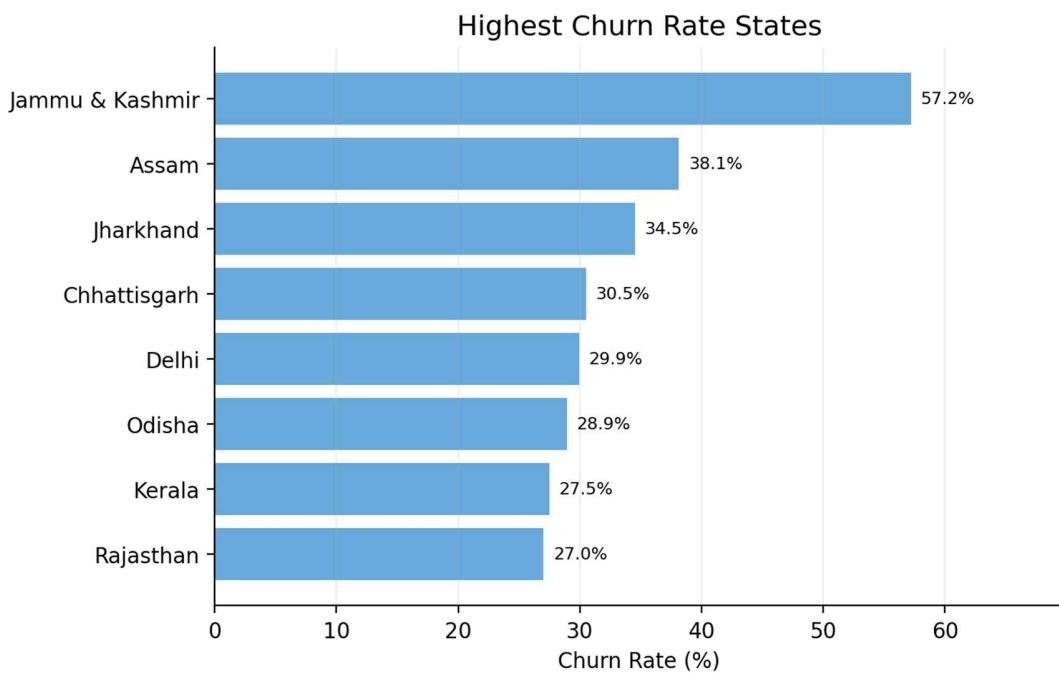


Figure 5. Highest churn-rate states.

Churn reasons indicate competitive pressure

Competitor-related churn is the largest churn category. This suggests that churn is not only a service-quality problem; it may also reflect competitor offers, devices, download speeds, and data plan differences.

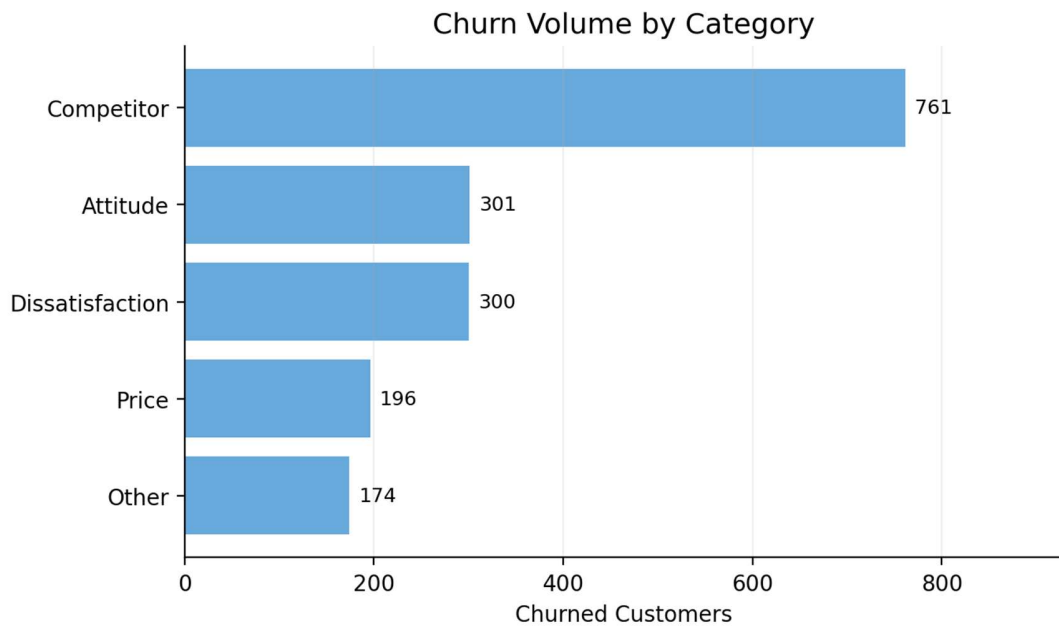


Figure 6. Churn volume by category.

7. Random Forest Churn Prediction Model

A Random Forest Classifier was trained to classify customers as stayed or churned. The notebook drops Customer_ID, Churn_Category, and Churn_Reason to avoid using identifiers or post-churn explanation fields as predictors. Categorical variables are label encoded and the target is mapped as Stayed = 0 and Churned = 1.

Model configuration

Item	Configuration
Algorithm	RandomForestClassifier
Estimators	100 trees
Train-test split	80/20
Random state	42
Target	Customer_Status
Target encoding	Stayed = 0, Churned = 1

Holdout performance from the notebook

Class	Precision	Recall	F1-score	Support
Stayed	0.85	0.93	0.89	862
Churned	0.78	0.59	0.67	340
Accuracy			0.84	1,202
Weighted average	0.83	0.84	0.83	1,202

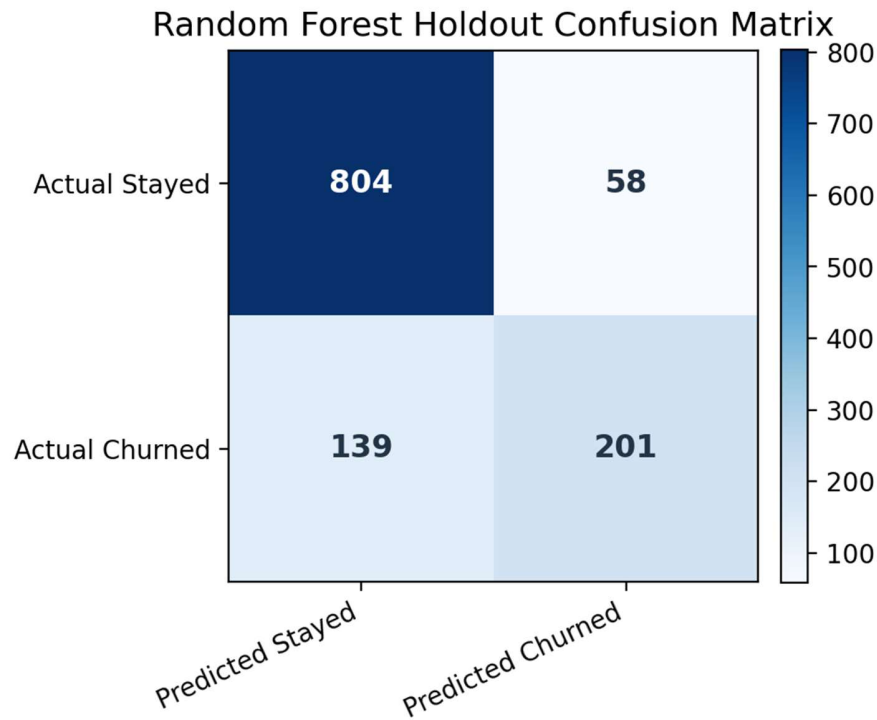


Figure 7. Holdout confusion matrix from the notebook output.

Model interpretation

The model performs well overall, but churn recall is 59%. For a retention use case, the next improvement should tune the classification threshold or class weights to catch more churners, even if precision decreases slightly.

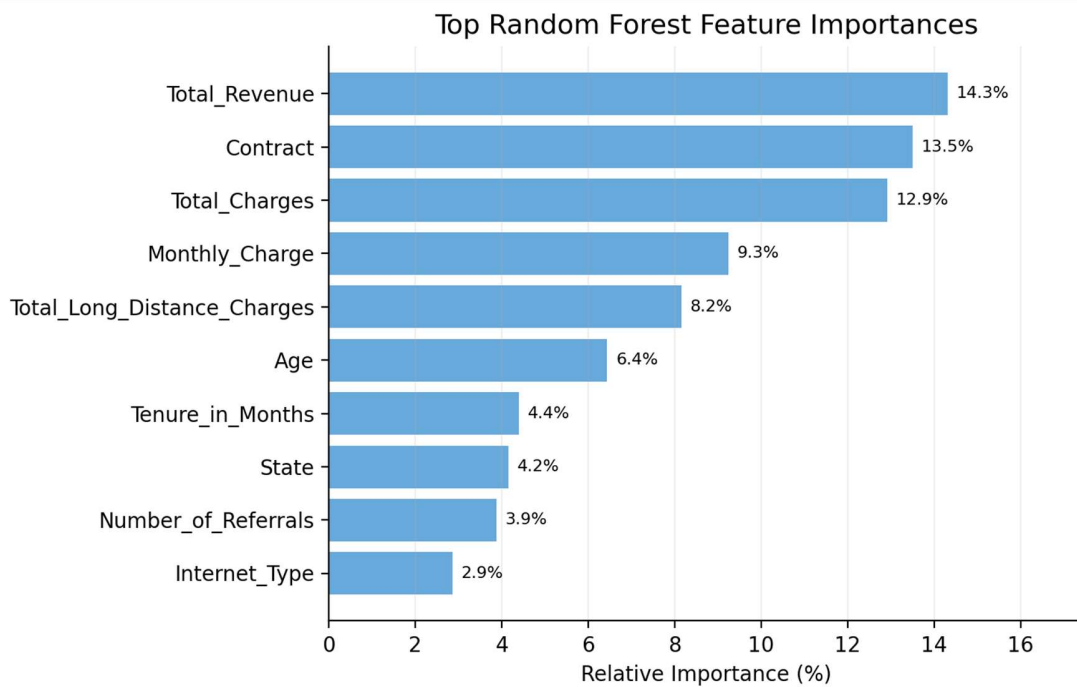


Figure 8. Top feature importances from a rerun on the provided CSV snapshot.

8. Predicted Churner Prioritization

The trained model was applied to new joiner data and produced 380 predicted churners. This output is valuable because it creates a practical retention worklist instead of only describing historical churn.

Predicted churner metric	Value
Predicted churners	380
Total monthly charge exposure	16,234.30
Total revenue exposure	43,121.35
Total referrals associated with predicted churners	2,775
Female / Male predicted churners	246 / 134
Month-to-month predicted churners	355

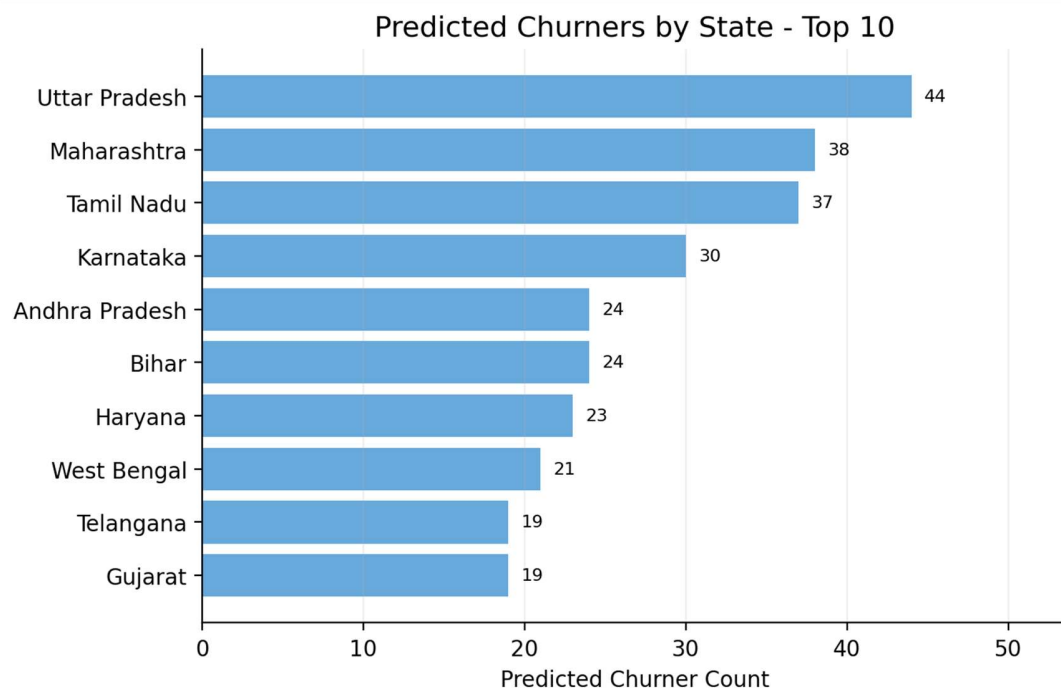


Figure 9. Predicted churners by state.

Recommended retention prioritization logic

- Prioritize predicted churners with high total revenue and high monthly charge.
- Within each state, rank predicted churners by value to assign outreach capacity fairly across regions.
- Treat month-to-month predicted churners as a special campaign audience for contract conversion.
- Use referrals as an additional prioritization factor because dissatisfied referred customers can influence broader acquisition quality.

9. Business Recommendations

Recommendation	Why it matters	Suggested action
Convert month-to-month customers	Highest visible contract churn risk	Offer loyalty discounts, annual plans, or bundled benefits

Launch regional retention campaigns	Churn is geographically uneven	Create state-level churn monitoring and regional playbooks
Respond to competitor churn	Competitor is the largest churn category	Review device offers, speed tiers, data plans, and pricing
Improve onboarding for new joiners	380 joiners are predicted churners	Trigger early-life support and plan education workflows
Optimize model for recall	Missed churners reduce retention opportunity	Tune threshold, use class weights, and track lift by decile

30-60-90 day implementation roadmap

Timeframe	Focus	Deliverable
First 30 days	Retention targeting	Top predicted churner call list by state and contract
Next 60 days	Offer testing	A/B test incentives for month-to-month and high-value customers
Next 90 days	Closed-loop learning	Campaign response table connected to SQL, Power BI, and model retraining

12. Appendix: SQL Business Questions

SQL section	Business question	Techniques used
0	How is the data cleaned and feature engineered?	CTEs, CASE logic, TRY_CONVERT, flags
1	What is the executive KPI snapshot?	CTE, window share calculations
2	Which segments have the highest churn rate and lift?	UNION ALL, DENSE_RANK, window lift
3	Which states deserve retention focus?	RANK, NTILE, cumulative SUM OVER
4	Which age and tenure cohorts are vulnerable?	Cohort aggregation, window averages
5	What churn reasons drive the Pareto impact?	ROW_NUMBER, cumulative SUM OVER
6	Which service subscriptions show elevated churn?	CROSS APPLY unpivot, service comparisons
7	Does monthly charge level change churn risk?	NTILE quintiles, contribution analysis
8	Which predicted churners should be called first?	ROW_NUMBER, NTILE, priority score
9	What are the highest-risk predicted segments?	DENSE_RANK, segmentation
10	Are there data-quality issues?	Issue CTE, RANK, issue share